

Portfolio Risk Return Balance

Recognition of Risk

- **Business Risk:** Ability of a firm to earn operational revenue is represented by the business risk. When a firm fails to earn through its operations, it leads to the erosion of capital contributed, and there is a risk of business failure. In this situation, the investor will not get any value from the share of the firm.
- **Market Risk:** Firm may have the ability to earn business profits, but the overall economic situation may force the market to value the share at a lesser value. This risk of going down with the market movements is known as market risk.

Recognition of Risk

- **Interest Rate Risk:** Value of debt securities varies inversely with interest rates. When interest rates are higher, the value of debt will be lower; while lower interest rates will push up the value of debt securities. Value of other types of securities may also be affected due to the rise or fall in interest rates.
- **Inflation Risk:** Inflation in the economy also influences the risk of investments. It may also result in the return from investments not matching the rate of increase in general price level (inflation). Change in the inflation rate also changes the consumption pattern, and hence investment returns carry an additional risk.

Recognition of Risk

- **Currency Risk:** Foreign holdings and international trade movements cause a change in the value of currency exchange rates. Currency risk is applicable mainly to firms that operate globally.
- **Political Risk:** Possibility that the government may change and hence the policies of the government may affect the business prospects. Policy changes in the tax structure, concessions and levy of duties to products, relaxation or tightening of foreign trade relations, etc. carry a risk component that will change the return pattern of businesses.

Firms' View of Risk

- Risk is commonly associated with the loss of the initial investment. It is referred to as the erosion of capital.
- Firms value the market price of the share as their capital worth. Fluctuations in the market price do not necessarily result in an erosion of capital, since the market continues to value the share even after a slump. However, when a firm is liquidated, the share value may come to a halt as on that date. Unless the liquidation value of the share is zero, the investor receives a certain amount for the investment.
- Investors in the capital market, need not necessarily lose fully capital unless they get out of the market in a slump or the firm goes into liquidation.

Portfolio Risk

- Market risk
 - Risk of all securities represented in a market
- Diversified risk
 - Diversification ensures that the return from one industry compensates for the loss if any that may occur due to the holding of a security representing another industry
- Risk adjustment economic valuation
 - Changes in risk events resulting in a favourable or unfavourable change in the economic value of asset exposures in the portfolio.

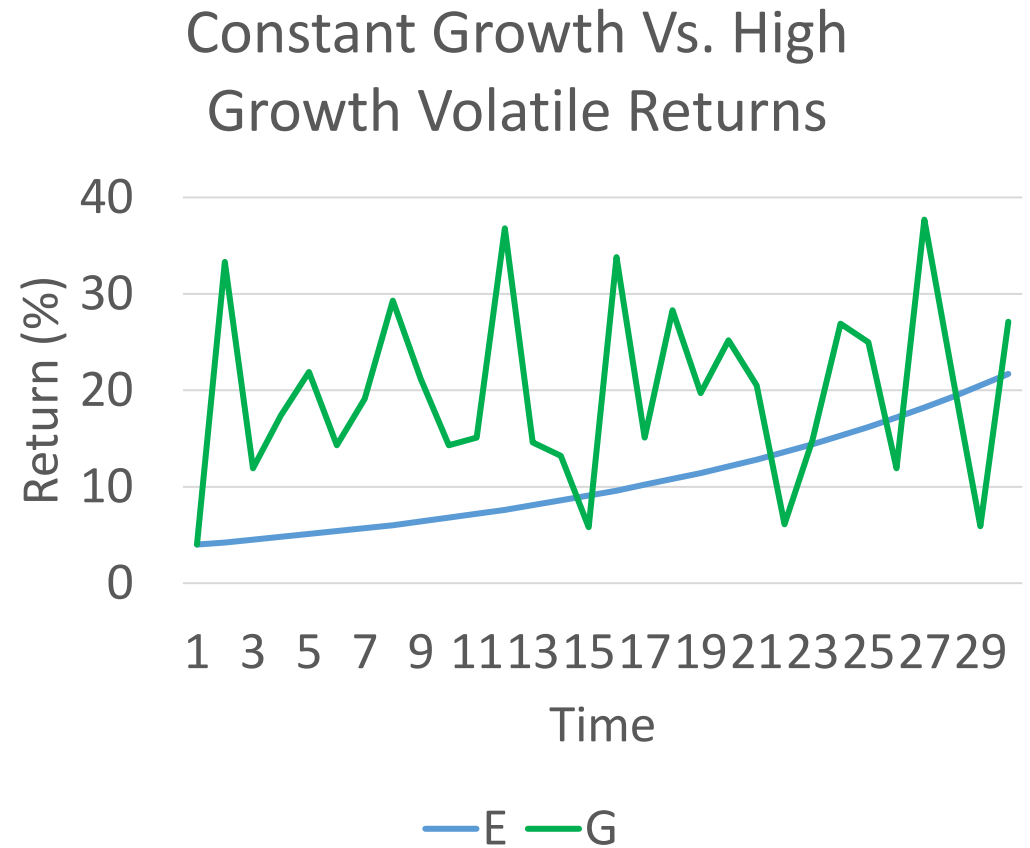
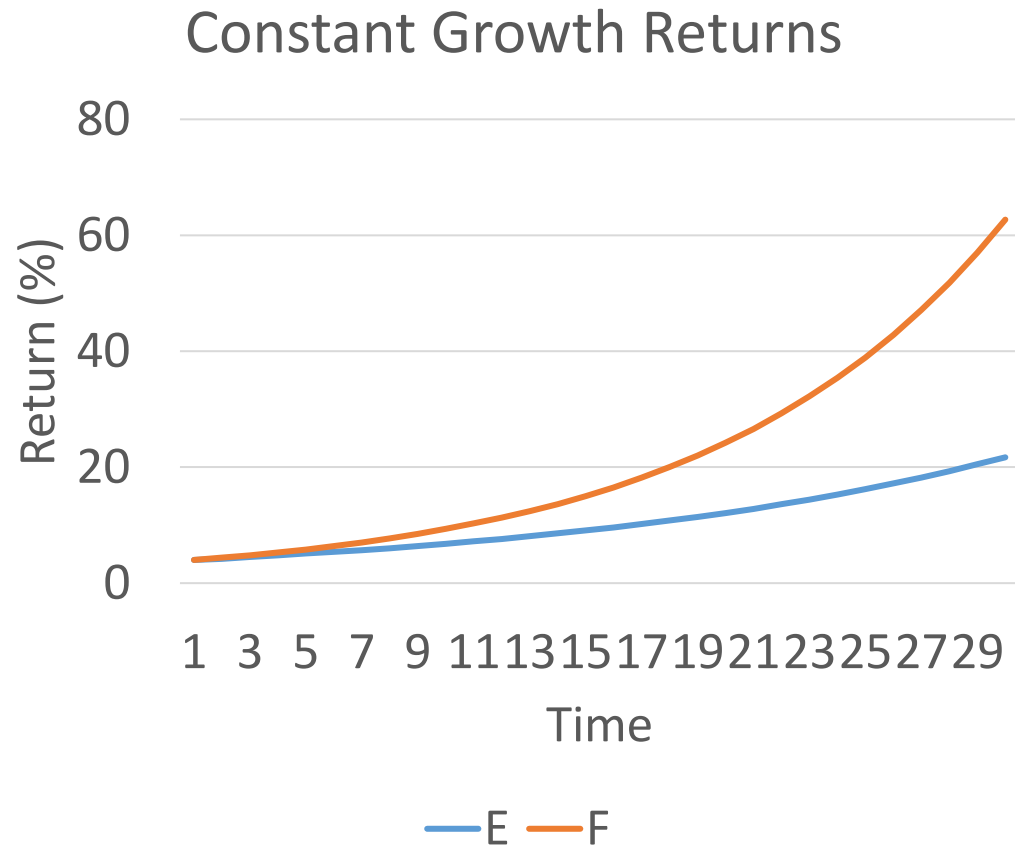
Characteristics of Securities

- Return
 - Change in value of security over time (day / month / year / 5 year)
- Risk
 - Risk is explained theoretically as the fluctuation in returns from a security.
- Safety
 - Assured return from securities over time
- Liquidity
 - Ability to convert security to cash at any time

Objectives of Investment

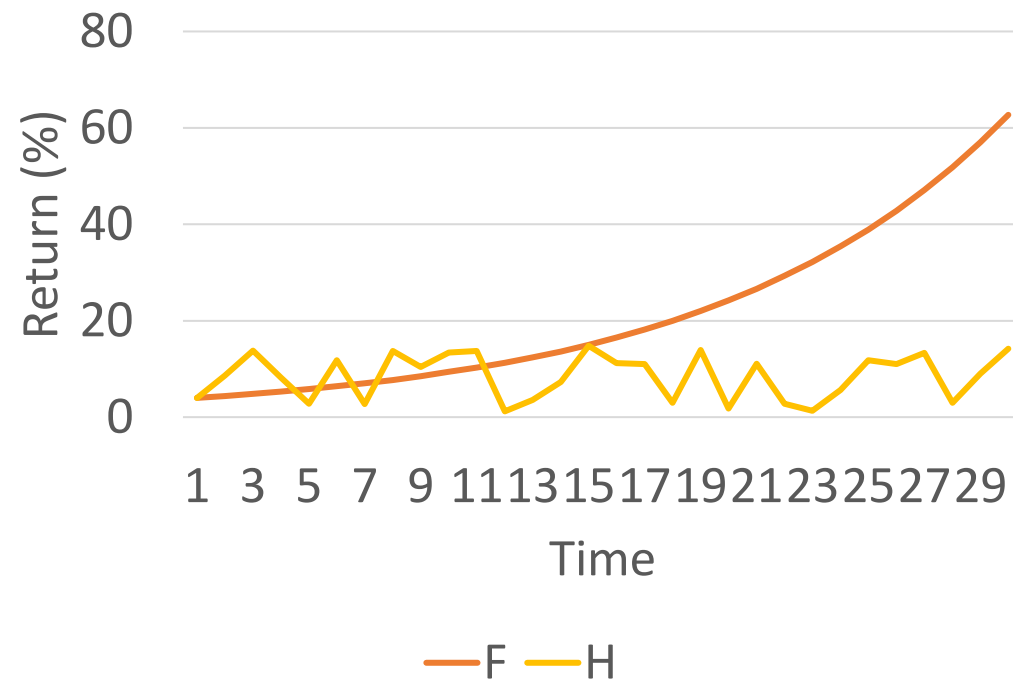
- Risk – Return balance
 - Preference for mutual funds if it offers higher than market return with moderate risk
 - Investment in IPOs when risk is perceived to be high would be made only when return expectation is exceptionally high
- Safety and Liquidity
 - Income yielding assets than growth yielding assets would be preferred by retired employees
 - Expected return from a financial asset is higher than the prevalent inflation rate
- Utilization of tax incentives
 - Reduce present tax payment and achieve a marginal rate of return on investments
 - Preference for provident fund / insurance schemes to reduce tax

Risk Return Dilemmas

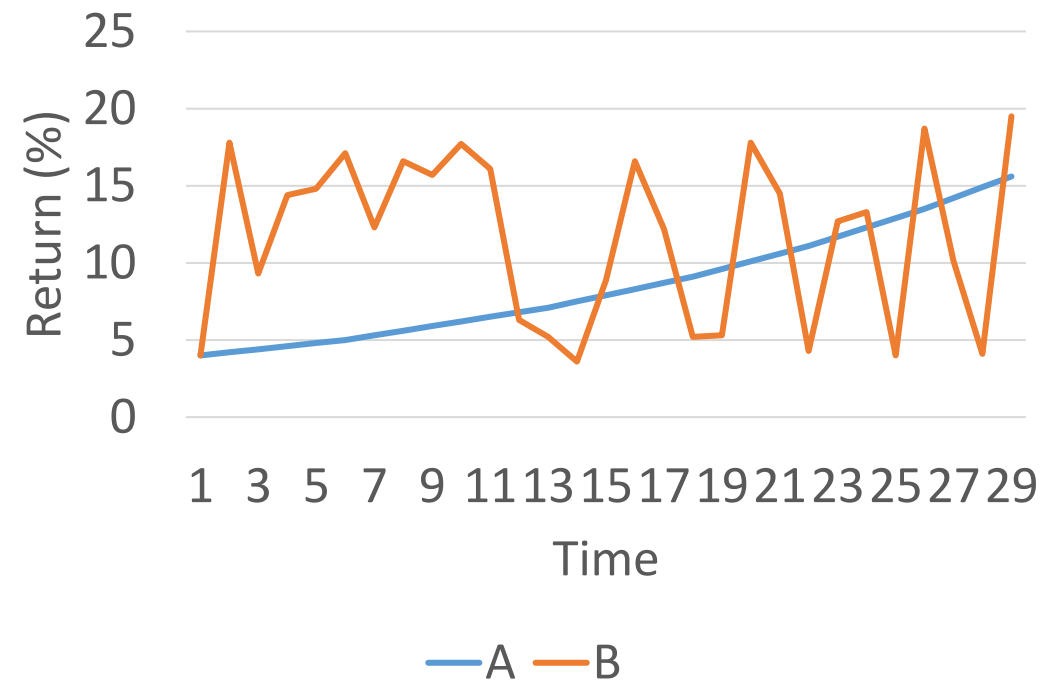


Risk Return Dilemmas

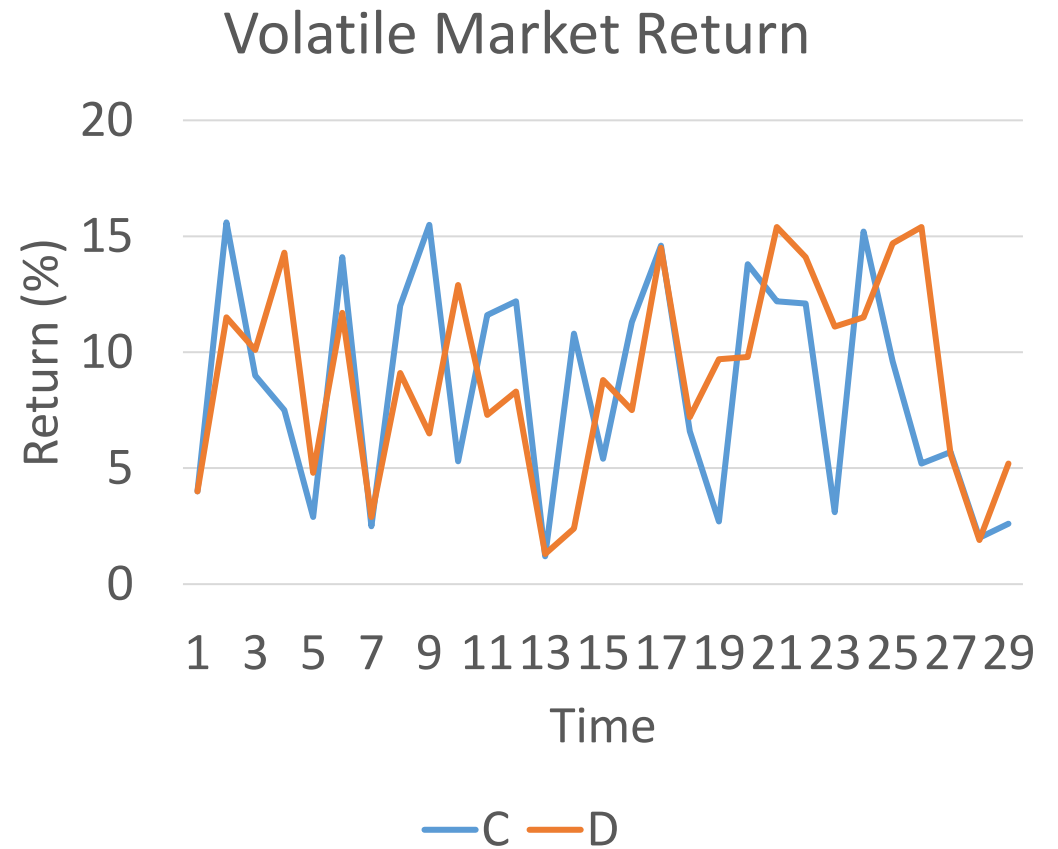
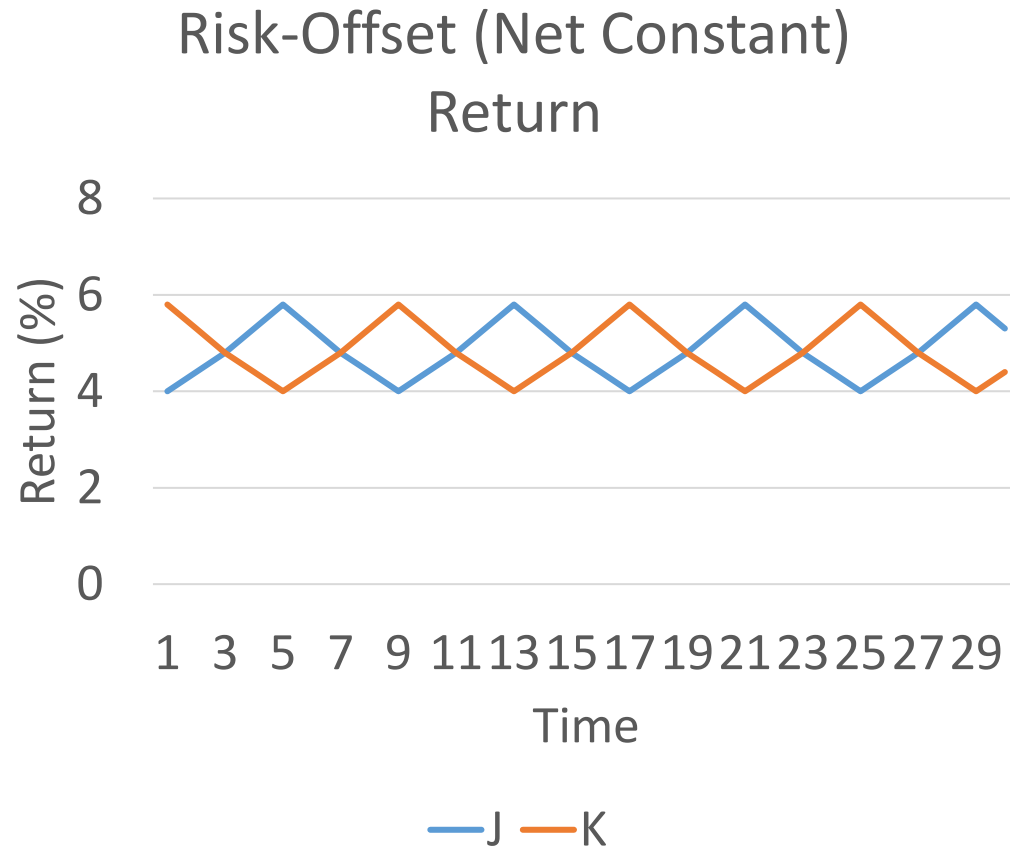
Constant Growth Vs. Low Growth Volatile Returns



Consistent Growth Vs. Volatile Returns



Risk Return Dilemmas



Systematic Risk Measure

- Total risk (variance) can be divided into the component that is unique to the security and that part that is associated with the market.
- Unique risk to the security is due to certain internal factors that are borne only by that firm
 - Risk due to poor management, risk due to labour unrest, poor product quality and so on.
- Unique risk can be reduced by diversification, and hence it is called diversifiable risk or unsystematic risk.
- Market risk relates to the risk that is inherent in the market and hence cannot be diversified.
- Market risk hence is termed as undiversifiable risk or systematic risk.

Statistical Measure of Components of Market Risk

- Total risk (variance) = Systematic risk + Unsystematic risk
- Total risk (variance) = Undiversifiable risk (Systematic) + Diversifiable (Unsystematic)
- Market movement can be represented through the market index, and all securities that are traded in the market are expected to have a positive relationship with market movements.
- Beta measures this relative change in security prices with respect to the change in the market movements.
- Beta is a measure of comparative risk between securities.

Measuring Security Beta

- Beta is the slope of the regression line modeled with market returns and security returns.
- $r_s = \alpha + (\beta \times r_m)$
- Where: r_s : Security return; r_m : Market return; α, β represent the intercept and slope of the regression line respectively.
- Beta can also be computed using the statistical measures of co-variance and market variance.
- **Security Beta = Co-variance/Market variance**
- Security Beta = [Correlation between security and market return $\times$ security standard deviation $\times$ market standard deviation] / Market variance.

Investor Inference of Beta

- Beta value less than one indicates a defensive share, as the return from the share will be less than that of the market return.
 - When the market moves up(down) by 10 per cent and the beta of the share is 0.50. the share return will move up(down) by only 5 per cent
- Beta value more than one indicates an aggressive security, as the return from the security will be more than that of the market return.
 - Beta of 2.5 implies, if market return moves up(down) by 10% the security return will move up(down) by 25%

High Beta Securities

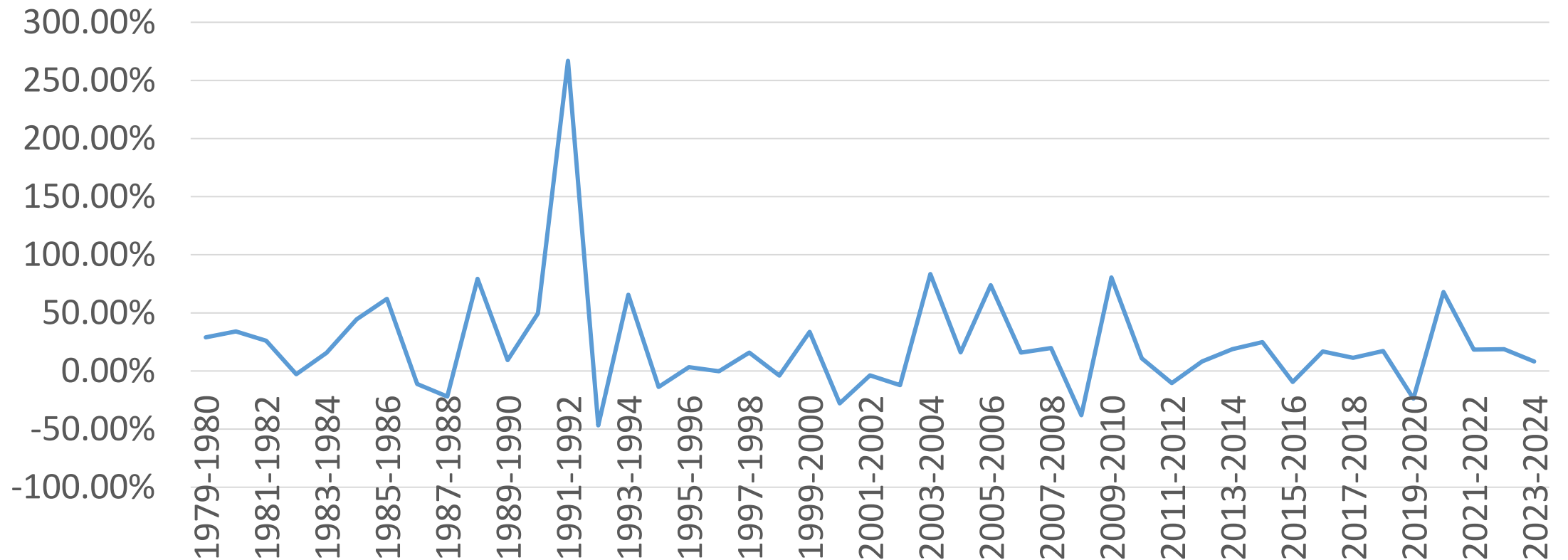
Name	Sub-Sector	Beta
Indusind Bank Ltd	Private Banks	2.33
Bajaj Finserv Ltd	Insurance	1.95
Bajaj Finance Ltd	Consumer Finance	1.89
Hindalco Industries Ltd	Metals – Aluminium	1.85
Tata Motors Ltd	Four Wheelers	1.73
Adani Enterprises Ltd	Commodities Trading	1.56
Axis Bank Ltd	Private Banks	1.54
Tata Steel Ltd	Iron & Steel	1.47
State Bank of India	Public Banks	1.44
ICICI Bank Ltd	Private Banks	1.32

Low Beta Securities

Name	Sub-Sector	Beta
Kansai Nerolac Paints Ltd.	Chemical	0.01
Gujarat Gas Company Ltd.	Energy	0.02
Strides Arcolab Ltd.	Pharma	0.03
Godfrey Philips India Ltd.	Tobacco	0.03
Kemrock Industries & Exports Ltd.	Plastic Products	0.05
Marico Ltd.	Edible oil	0.06
Gulf Oil Corporation Ltd.	Energy	0.06
HEG Ltd.	Electrodes	0.08
Cadila Healthcare Ltd.	Pharma	0.09
Page Industries Ltd.	Textile	0.1

Market Return

Sensex Annual Returns



Measurement of Systematic Risk

- Market Return: 2023-24 : 8.17%
- Systematic risk = beta x market return
- Tata Motors: $1.73 \times .0817 = 14.13\%$

Name	Beta	Systematic Risk
Tata Motors	1.73	14.13%
Adani Enterprises	1.56	12.75%
Gujarat Gas	0.02	0.16%
Marico	0.06	0.49%

Measurement of Unsystematic Risk

- Total risk (variance) = Systematic risk + Unsystematic risk
- Total Risk $\sigma^2 = (\beta \times \sigma_m^2) + \sigma_{ei}^2$
- Unsystematic risk is firm specific unique risk that is segregated from the total price variations measured in the market.
- Unsystematic risk variance $\sigma_{ei}^2 = \sigma^2 - (\beta \times \sigma_m^2)$

Measurement of Unsystematic Risk

- Total Risk: Standard deviation

Name	Systematic Risk	Volatility (Standard Deviation)
Tata Motors	14.13%	1.75%
Adani Enterprises	12.75%	2.19%
Gujarat Gas	0.16%	0.49%
Marico	0.49%	0.54%

Measurement of Unsystematic Risk

Name	Total Risk: Variance (%)	Systematic Risk : Variance (%)	Firm Specific Risk: Variance (%)	Firm Specific Risk : Standard Deviation (%)
Tata Motors	3.50%	1.15%	2.34%	15.30%
Adani Enterprises	4.38%	1.04%	3.34%	18.26%
Gujarat Gas	0.98%	0.01%	0.97%	9.85%
Marico	1.09%	0.04%	1.05%	10.23%

Category of Market Risk

